

Akhil Ganji

Software Development Engineer

+1 (682) 231- 3650 | [Email](#) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Professional Summary

Software Development Engineer with over 5 years of experience designing, developing, and optimizing enterprise-scale applications across healthcare, financial, and IT domains. Proficient in **Java, Python, C++, Node.js, and React.js**, with strong expertise in **microservices architecture, event-driven systems, and cloud-based solutions** (AWS, Azure, Docker, Kubernetes). Experienced in building **high-performance backend systems, ETL pipelines, and database optimizations** using PostgreSQL, MySQL, MongoDB, and Redis, achieving measurable improvements in throughput, latency, and reliability. Adept at implementing **CI/CD pipelines, automated testing, and observability frameworks** (JUnit, Mockito, TestContainers, ELK Stack, OpenSearch) to enhance deployment stability and operational efficiency. Skilled in **API security, OAuth2, JWT, RBAC**, and integrating **LLM-powered workflows** to automate complex processes. Strong collaborator in Agile/Scrum environments, working effectively with **ML, DevOps, and frontend teams** to deliver scalable, maintainable, and business-aligned software solutions.

Technical Skills

Programming Languages: Java 8/11, Python, JavaScript, TypeScript, C++, C#

Frameworks & Libraries: Spring Boot, Hibernate, Django, FastAPI, Node.js, Express, React.js, REST, gRPC, Kafka, RabbitMQ

Databases & Data Management: PostgreSQL, MySQL, MongoDB, Redis, ETL pipelines, Pandas, NumPy, Data Validation, Indexing, Query Optimization

Cloud Platforms & DevOps Tools: AWS (Lambda, RDS, CloudWatch, S3), Azure, Docker, Kubernetes, Jenkins, GitHub Actions, Terraform, ELK Stack, OpenSearch

Testing & QA: JUnit, Mockito, TestContainers, Selenium, Unit Testing, Integration Testing, Functional Testing, Code Reviews

Security & Access Management: OAuth2, JWT, RBAC, API Security, Enterprise Compliance (HIPAA-aligned), Authentication & Authorization

Software Architecture & Design: Microservices Architecture, Event-driven & Asynchronous Systems, Modular Design, OOP, SOLID Principles, High-Performance System Optimization

AI / Machine Learning Integration: LLM Integration (GPT, embeddings, inference), Intelligent Workflow Automation, Prompt Engineering, Cross-team Collaboration (ML, DevOps, Product Teams)

Work Experience

Cigna Healthcare, USA | Software Engineer

Apr 2025 – Present

- Engineered scalable **Java 17 & Spring Boot microservices**, improving healthcare claim processing throughput by 30% through robust OOP, DSA optimization, and concurrency control.
- Integrated **LLM-powered automation workflows** using GPT inference, embeddings, and prompt engineering, cutting manual claim validation time by 25%.
- Optimized backend and API performance using **Hibernate, Redis caching, and async processing**, reducing latency by 35% and improving real-time transaction efficiency.
- Implemented enterprise-grade **logging, monitoring, and alerting** with CloudWatch, OpenSearch, and ELK Stack, decreasing incident resolution time by 20%.
- Accelerated deployments by 50% through robust **CI/CD pipelines using Jenkins, GitHub Actions, Docker, and Kubernetes**.
- Lifted test coverage to 85% using **JUnit, Mockito, and TestContainers**, performing code reviews and collaborating effectively across ML, DevOps, and product teams in Agile/Scrum.

Cognizant Inc, India | Software Engineer

Mar 2020 – Aug 2023

- Developed **distributed backend systems** using Python, Django, and FastAPI, optimizing architecture and algorithms to enhance **transaction throughput by 20%** for enterprise-scale applications.
- Built **high-volume ETL pipelines** with Python, Pandas, and NumPy, reducing data processing time by 15% while ensuring **high reliability, accuracy, and maintainability** of enterprise datasets.
- Designed and optimized **relational database schemas** in PostgreSQL/MySQL, implementing indexing strategies and **Redis caching** to reduce query latency by 35% and improve system responsiveness.
- Implemented **event-driven microservices** using Kafka and RabbitMQ, improving system **reliability, scalability, and fault tolerance** across distributed applications.
- Secured APIs using **OAuth2, JWT, and RBAC**, ensuring enterprise-level security compliance and protecting sensitive financial and client data.
- Added **structured logging, monitoring, and error-handling mechanisms**, enhancing system observability, debugging efficiency, and operational insights for production environments.
- Conducted **code reviews**, collaborated in Agile sprints, and coordinated integrations with frontend and DevOps teams to ensure smooth development and deployment cycles.
- Documented APIs, workflows, and design decisions to support **long-term maintainability, team onboarding, and**

knowledge transfer across the organization.

DXC Technologies, India | Software Engineer

Mar 2019 – Feb 2020

- Contributed to **enterprise application modules** using Java, Python, and C++, applying **OOP, data structures, algorithms, and clean code principles** to deliver robust and maintainable software solutions.
- Assisted in developing **REST, gRPC, and Node.js APIs** to enable seamless communication between internal systems, improving **response consistency and system interoperability**.
- Enhanced release stability by **35%** through comprehensive **unit and integration testing** using JUnit, Mockito, and Selenium, ensuring reliable and smooth deployment cycles.
- Supported backend development tasks involving **multithreading, concurrency, and event-driven patterns**, optimizing **system performance, throughput, and resource utilization**.
- Developed **frontend interfaces** using React.js, HTML, and CSS, and integrated **MongoDB** for internal tool data management, improving **usability, reporting efficiency, and data accessibility**.

Education

Master of Science, Information Technology

Aug 2025

Webster University, Saint Louis, Missouri, USA